

E-Commerce Microservices Platform – Project Report

Project Overview

Project Title: E-Commerce Microservices Platform

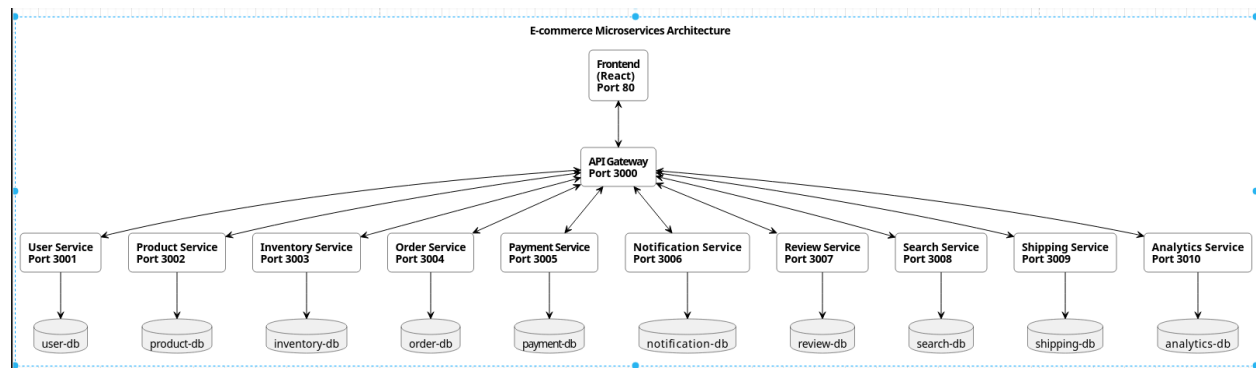
Technology Stack: MERN (MongoDB, Express.js, React.js, Node.js)

Architecture: Microservices with Docker Containerization

Total Components: 12 Microservices + 1 Frontend + 1 API Gateway

Total Containers: 22 Docker Containers

System Architecture



The platform follows a **microservices-based architecture**, structured into multiple independent layers:

- **Client Layer:** React.js frontend interface
- **API Gateway Layer:** Unified entry point managing routing and authentication
- **Microservices Layer:** 12 independent business services
- **Database Layer:** 10 dedicated MongoDB instances (one per service)
- **Infrastructure Layer:** All components containerized and orchestrated via Docker Compose

Screenshots Documentation

1 All Containers Running

```
saadkhan@saadkhan-HP-ZBook-14u-G5:~/Desktop/e-commerce$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
61414e385e76	e-commerce_frontend	"/docker-entrypoint..."	13 minutes ago	Up 13 minutes	0.0.0.0:80->80/tcp, [::]:80->80/tcp	frontend
87cbd8759cd1	e-commerce_api-gateway	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	api-gateway
be44d489fb11	e-commerce_shipping-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3009->3009/tcp, [::]:3009->3009/tcp	shipping-service
bfa1e4fbf4	e-commerce_product-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3002->3002/tcp, [::]:3002->3002/tcp	product-service
bb33ecbf9621	e-commerce_notification-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3006->3006/tcp, [::]:3006->3006/tcp	notification-service
4ad527e858f6	e-commerce_analytics-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3010->3010/tcp, [::]:3010->3010/tcp	analytics-service
84d7aa4e2ca3	e-commerce_inventory-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3003->3003/tcp, [::]:3003->3003/tcp	inventory-service
0f628e16a352	e-commerce_payment-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3005->3005/tcp, [::]:3005->3005/tcp	payment-service
7bd2867056e	e-commerce_user-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp	user-service
42828dd487e3	e-commerce_review-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3007->3007/tcp, [::]:3007->3007/tcp	review-service
ef78b1c6deb1	e-commerce_order-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3004->3004/tcp, [::]:3004->3004/tcp	order-service
4ffd95c49d34	e-commerce_search-service	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:3008->3008/tcp, [::]:3008->3008/tcp	search-service
e6b37c3899fd	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27018->27017/tcp, [::]:27018->27017/tcp	product-db
6424b8026eb3	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27025->27017/tcp, [::]:27025->27017/tcp	shipping-db
58a85d7f249f	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp	user-db
25d4d374fa1f	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27021->27017/tcp, [::]:27021->27017/tcp	payment-db
80a0f468224b	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27023->27017/tcp, [::]:27023->27017/tcp	review-db
363273f73b98	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27026->27017/tcp, [::]:27026->27017/tcp	analytics-db
825a6aa7946c	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27020->27017/tcp, [::]:27020->27017/tcp	order-db
03fb3f8ccbe1	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27022->27017/tcp, [::]:27022->27017/tcp	notification-db
94e7ebf259b1	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27019->27017/tcp, [::]:27019->27017/tcp	inventory-db
dd51bc6b40df	mongo:7.0	"docker-entrypoint.s..."	13 minutes ago	Up 13 minutes	0.0.0.0:27024->27017/tcp, [::]:27024->27017/tcp	search-db

```
saadkhan@saadkhan-HP-ZBook-14u-G5:~/Desktop/e-commerce$
```

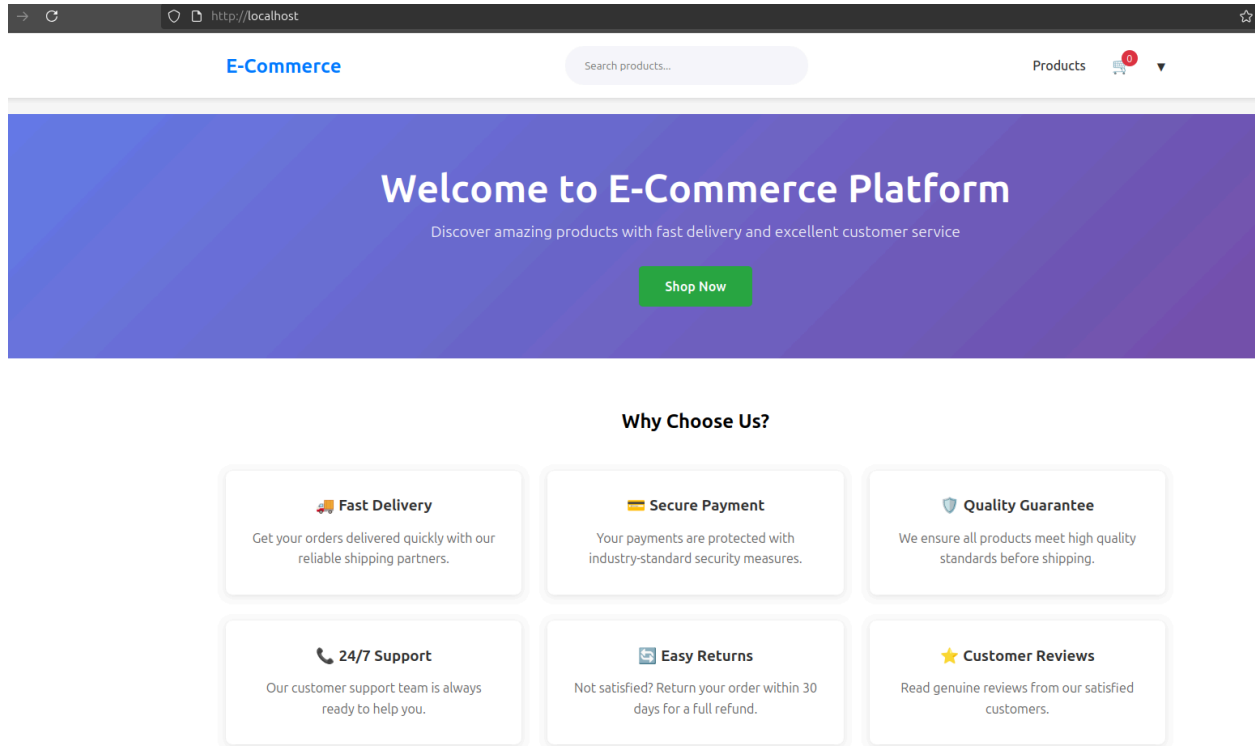
Description:

Displays all 22 containers running successfully:

- 12 microservices (ports 3001–3010)
- 10 MongoDB databases (ports 27017–27026)
- 1 React frontend (port 80)
- 1 API Gateway (port 3000)

Command Used: `docker-compose ps`

2 Main Application Homepage

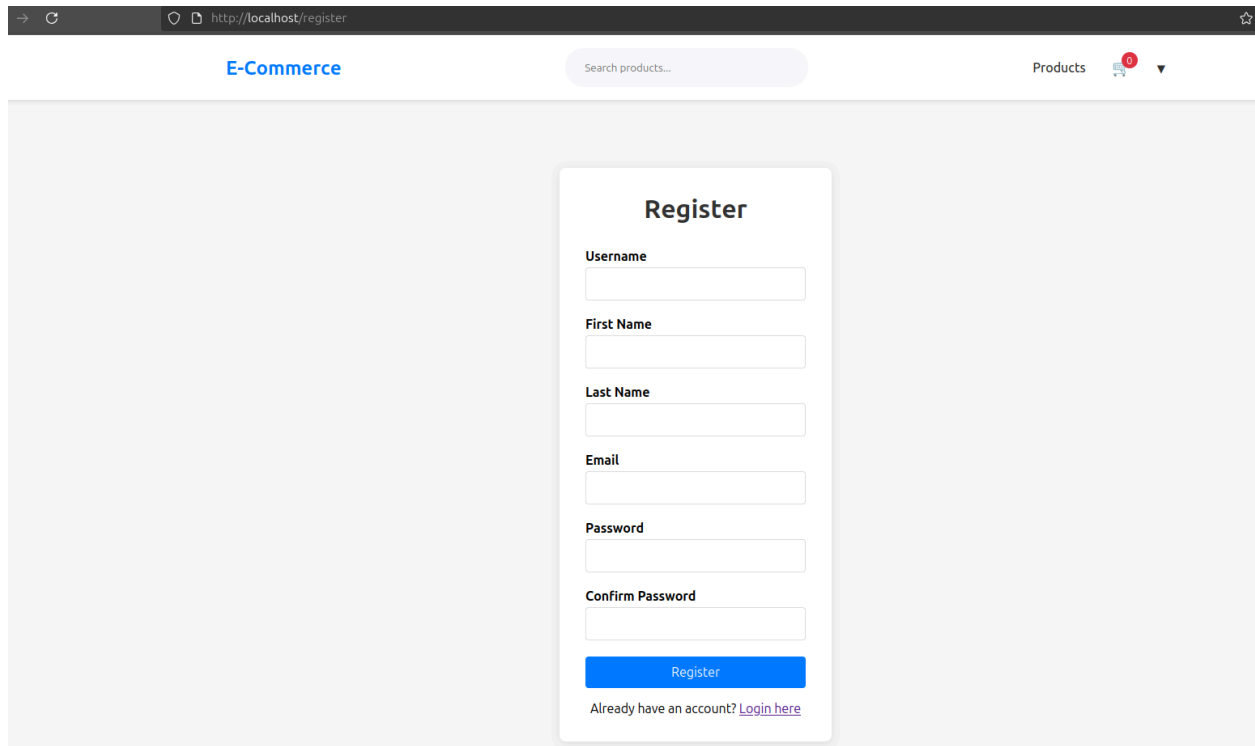


Description:

Homepage of the React frontend displaying navigation, hero section, and product listings.

URL: <http://localhost/>

3 User Registration Page



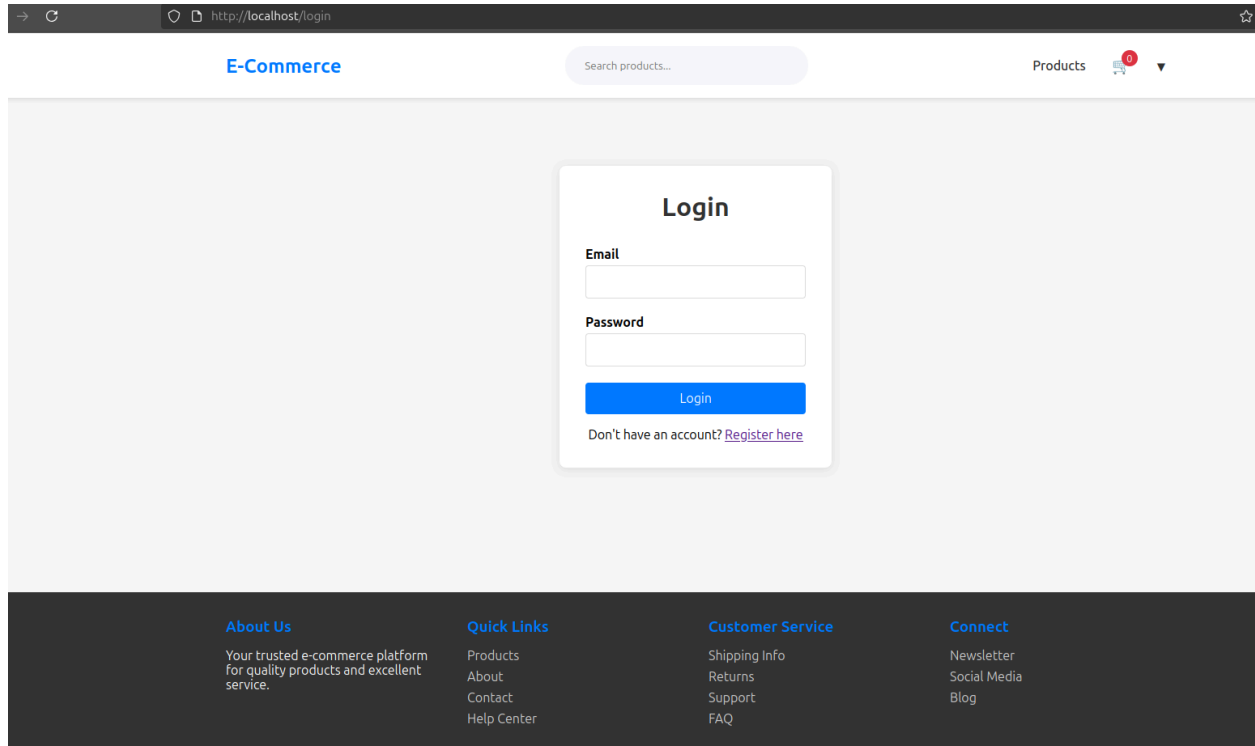
The screenshot shows a web browser at the URL `http://localhost/register`. The page features a header with the "E-Commerce" logo, a search bar with the placeholder "Search products...", and a "Products" link with a shopping cart icon and a red notification badge. The main content area is a light gray rectangle containing a white "Register" form. The form includes input fields for "Username", "First Name", "Last Name", "Email", "Password", and "Confirm Password". A blue "Register" button is positioned below the "Confirm Password" field. At the bottom of the form, there is a link that says "Already have an account? [Login here](#)".

Description:

Registration form allowing new users to create accounts.

Connected to the **User Service (Port 3001)** for user creation.

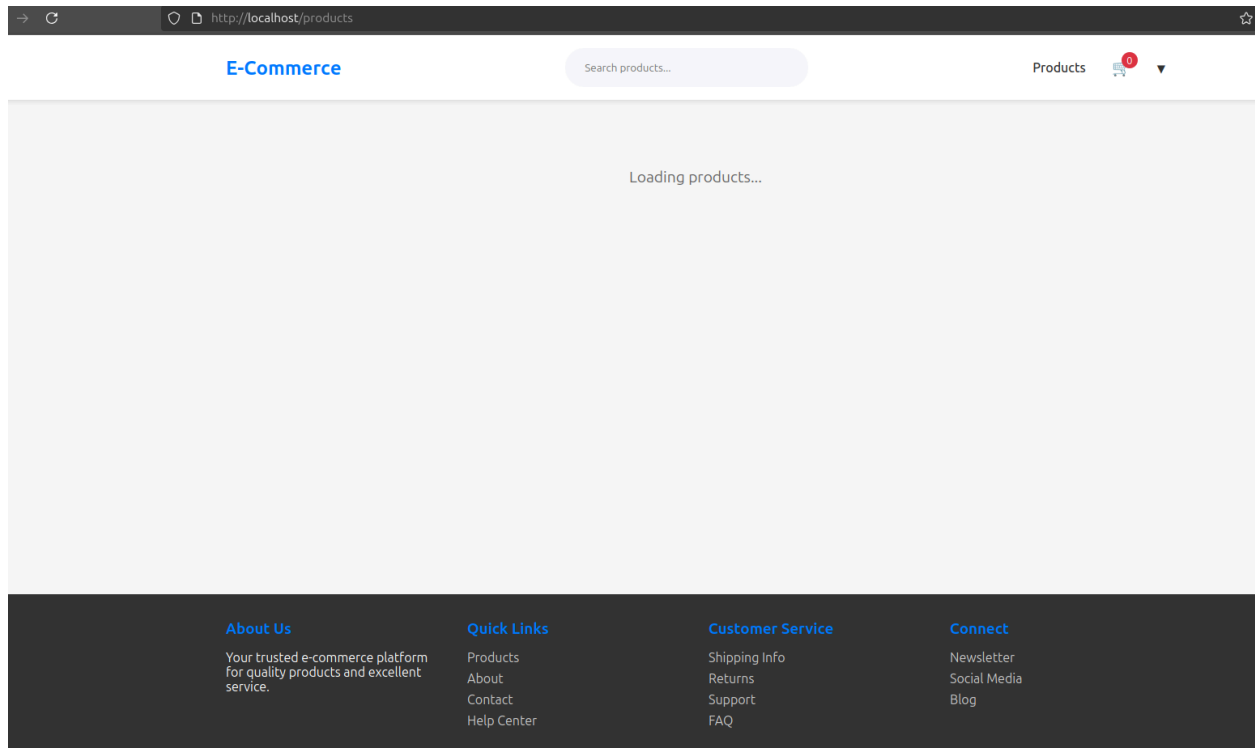
4 User Login Page



Description:

User login form authenticating via the **User Service**, returning **JWT tokens** for session management.

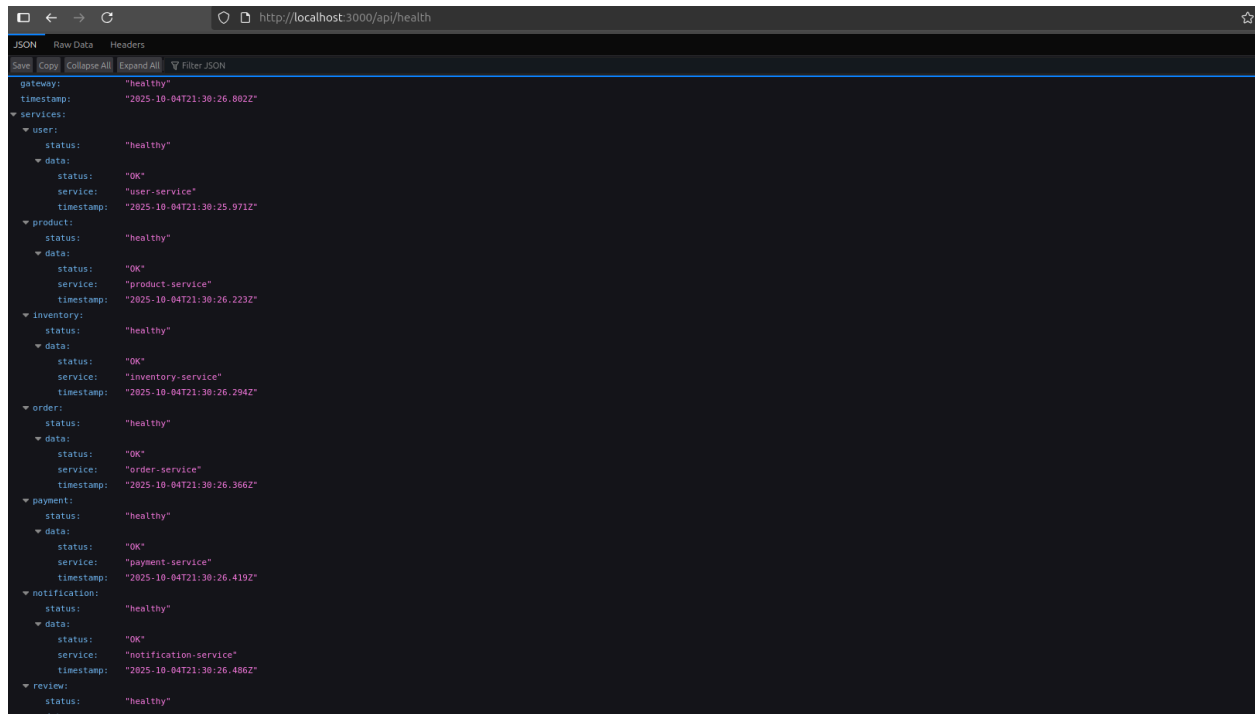
5 Products Page



Description:

Displays the product catalog fetched from the **Product Service (Port 3002)**.
Includes product images, names, prices, and details.

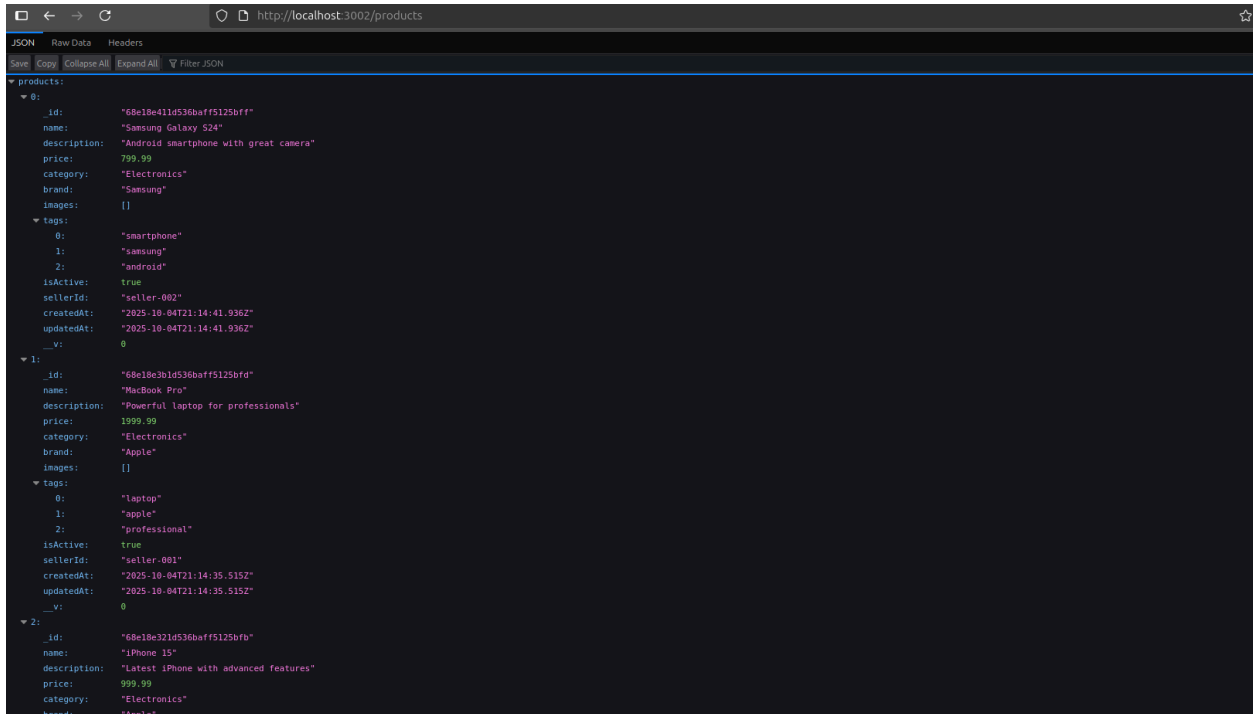
6 API Gateway Health Check



Description:

Shows a JSON health status of all microservices via the API Gateway.

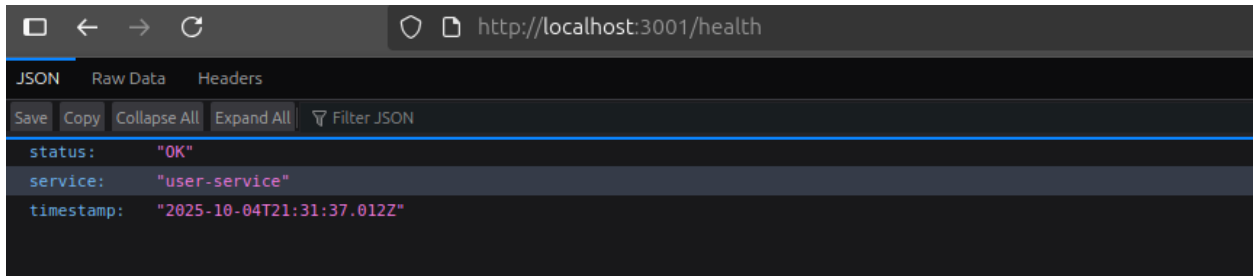
7 Product Service API Response



```
products:
  0:
    _id: "68e18e41d536baff5125bfff"
    name: "Samsung Galaxy S24"
    description: "Android smartphone with great camera"
    price: 799.99
    category: "Electronics"
    brand: "Samsung"
    images: []
    tags:
      0: "smartphone"
      1: "samsung"
      2: "android"
    isActive: true
    sellerId: "seller-002"
    createdAt: "2025-10-04T21:14:41.936Z"
    updatedAt: "2025-10-04T21:14:41.936Z"
    __v: 0
  1:
    _id: "68e18e3bd536baff5125bffd"
    name: "MacBook Pro"
    description: "Powerful laptop for professionals"
    price: 1999.99
    category: "Electronics"
    brand: "Apple"
    images: []
    tags:
      0: "Laptop"
      1: "apple"
      2: "professional"
    isActive: true
    sellerId: "seller-001"
    createdAt: "2025-10-04T21:14:35.515Z"
    updatedAt: "2025-10-04T21:14:35.515Z"
    __v: 0
  2:
    _id: "68e18e32d536baff5125bffb"
    name: "iPhone 15"
    description: "Latest iPhone with advanced features"
    price: 999.99
    category: "Electronics"
    brand: "Apple"
```

Description:
JSON response from the Product Service demonstrating product retrieval functionality.

8 User Service API Response



```
status: "OK"
service: "user-service"
timestamp: "2025-10-04T21:31:37.012Z"
```

Description:
JSON health response confirming the **User Service** is operational.

 Microservices Overview

Core Business Services

1. **User Service (Port 3001)**

- Handles registration, authentication, and JWT management
- Role-based access control
- Profile management

2. **Product Service (Port 3002)**

- Product catalog and category management
- Product CRUD operations
- Search and filtering

3. **Inventory Service (Port 3003)**

- Stock tracking and management
- Low-stock alerts and reservations

4. **Order Service (Port 3004)**

- Order creation, status tracking, and history
- Order cancellation

5. **Payment Service (Port 3005)**

- Payment processing and refunds
- Transaction logging

6. **Notification Service (Port 3006)**

- Email, SMS, and push notifications
- Custom notification templates

7. **Review Service (Port 3007)**

- Customer reviews and ratings

- Moderation and analytics

8. **Search Service (Port 3008)**

- Full-text search and auto-suggestions
- Search analytics and filters

9. **Shipping Service (Port 3009)**

- Delivery tracking and address validation
- Shipping cost calculation

10. **Analytics Service (Port 3010)**

- Business insights and user analytics
- Event and sales tracking

Infrastructure Services

11. **API Gateway (Port 3000)**

- Unified request routing
- Authentication and load balancing

12. **Frontend (Port 80)**

- React.js web interface
- Fully responsive and modern UI

Database Architecture

Each microservice maintains its **own MongoDB instance**:

Database Name	Port	Purpose
user-db	27017	User profiles
product-db	27018	Product data
inventory-db	27019	Stock management
order-db	27020	Orders and history
payment-db	27021	Transactions
notification-db	27022	Notifications
review-db	27023	Reviews and ratings
search-db	27024	Search indexes
shipping-db	27025	Delivery data
analytics-db	27026	Business metrics

Project Overview & Conclusion

The project has been successfully developed and deployed as a **fully containerized E-commerce Microservices Platform** featuring:

- **1 Frontend (React.js)** - Modern user interface
- **12 Microservices** - Independent and modular business services
- **10 Databases** - Dedicated MongoDB instance per service
- **22 Docker Containers** - Complete containerization setup
- **End-to-End E-commerce Flow** - From user registration to order completion

Deliverables Included

Source Code Structure - Organized and well-documented file hierarchy
Dockerfiles (12 Total) - Individual Dockerfiles for each microservice
Docker Compose Configuration - Single YAML file for orchestration
Package.json Files - Dependencies and scripts for each service